



Autonomous AI Bot for Self-Healing and Proactive Server Health Management in Distributed Systems

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Research Proposal

Autonomous AI Bot for Self-Healing and Proactive Server Health Management in Distributed Systems

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Declaration

We certify that this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any university and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where due reference is made in the text.

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1. Introduction and Background

1.1 Overview

1.1.1 Introduction

Distributed computing has become a vital part of modern digital infrastructures, supporting industries like healthcare, banking, telecommunications, and e-commerce (Wang et al., 2024). With the rapid growth of cloud-based services, the need for systems that are reliable, fault-tolerant, and resilient has never been higher. Traditional server monitoring relies heavily on human operators to spot issues and take corrective measures. However, as systems become larger and more complicated, and as data volumes increase dramatically, these manual methods lead to slower responses, higher operational costs, and an increased risk of downtime (Zhang & Wu, 2024).

Self-healing systems introduce automation into this important area. They allow infrastructures to detect faults, figure out their causes, and automatically take corrective actions. Artificial Intelligence (AI) further improves this idea by enabling predictive maintenance, more precise anomaly detection, and the creation of understandable decision-making processes that support human oversight (Das & Roy, 2024). This research builds on these advancements by suggesting the creation of an autonomous AI bot for self-healing in distributed systems. It aims to tackle key limitations of current monitoring solutions that still rely on manual input and have limited flexibility.

1.1.2 Background

The idea of self-healing computing dates back to IBM's Autonomic Computing Initiative in the early 2000s. This initiative outlined a vision of systems capable of self-configuration, self-healing, self-optimization, and self-protection (Kumar & Gupta, 2024). Over the past two decades, improvements in machine learning (ML) and reinforcement learning (RL) have made it possible to create systems that can recover from failures by adapting their actions based on feedback from their environment (Li et al., 2025). Despite this progress, modern infrastructures like Kubernetes, Docker Swarm, and multi-cloud platforms face significant challenges. Notably, the failure of a single node can lead to widespread service disruptions due to interdependent services (Müller & Hoffmann, 2024).

The financial and operational impacts of such failures are serious. According to Gartner (2024), downtime costs organizations between USD 300,000 and 500,000 per hour, with even greater losses in critical fields like finance and healthcare. These numbers stress the urgent need for autonomous fault management. Nevertheless, existing solutions struggle with ongoing issues. Anomaly detection systems often produce high false positive rates, leading to unnecessary and expensive recovery actions (Qureshi & Malik, 2025). Many AI driven systems lack transparency, which makes administrators reluctant to trust their automated decisions (Das & Roy, 2024). Additionally, compatibility issues with older infrastructures limit the scalability and deployment of fully autonomous healing agents (Brown & Green, 2025).

This research suggests a hybrid approach to address these challenges. The framework combines anomaly detection using ML models with healing strategies based on reinforcement learning. It also includes explainable AI to ensure human administrators can understand and validate system actions. By improving accuracy, adaptability, and transparency, the proposed system aims to enhance resilience and operational trust. This represents a meaningful step forward in the development of AI driven distributed systems.

1.2 Project Scope

The project aims to design, develop, and evaluate an AI based bot that autonomously manages server health in a distributed cloud environment.

- The system will focus on detecting anomalies in real time using log and performance data.
- It will use reinforcement learning to decide on suitable actions for healing, such as restarting containers, reallocating workloads, and blocking malicious IPs.
- An explainability engine will provide clear justifications, so administrators understand and trust the AI's decisions.
- The scope includes implementing a prototype environment with Docker and Kubernetes clusters to simulate distributed systems.
- The evaluation will look at detection accuracy, false positives and negatives, scalability, and mean time to heal (MTTH) compared to baseline monitoring solutions. The project will not cover full commercial deployment but will discuss the feasibility of commercialization.

1.3 Motivation for the Study

The main motivation comes from the growing complexity of distributed systems and the high cost of downtime. Studies estimate that just a few minutes of server downtime can result in financial losses from thousands to millions of dollars, depending on the industry (Rehman & Adrian, 2025; Sanusi et al., 2025).

Current methods still require human administrators to investigate anomalies. This leads to delays and increases the chance of human error. With the rapid rise in cyberattacks, like DDoS and ransomware, and the expansion of microservices-based deployments, there is a strong need for systems that can self-repair without constant supervision (Martin, 2024; Qureshi & Malik, 2025).

Furthermore, while AI has shown promise in spotting anomalies, administrators often distrust "black box" models. This creates a need for explainable AI (XAI), which supports adoption by providing transparency and accountability (Das & Roy, 2024; Park & Kim, 2025).

1.4 Rationale and Justification

This study is important for both technical and practical reasons. It aims to improve system management practices while also tackling real world industry challenges.

Technical Rationale

Traditional monitoring systems for distributed networks depend heavily on human operators to find failures and start manual recovery. This reactive method leads to slow responses, a greater chance of human mistakes, and limits scalability in large and changing settings like enterprise data centers and cloud infrastructures (Shevchenko, 2025). The proposed research shifts from this reactive approach to a proactive and autonomous system management model. By

combining AI based anomaly detection with reinforcement learning driven recovery strategies, the system aims to predict failures before they escalate and take quicker recovery actions. Adding an explainability layer ensures that administrators receive clear reasons for every automated decision, which promotes transparency and builds trust in the use of AI technologies (Das & Roy, 2024).

Practical (Industrial) Rationale

From an industrial perspective, this research directly targets the financial and operational risks associated with downtime and service interruptions. Studies reveal that unexpected outages can cost organizations hundreds of thousands of dollars each hour, with even larger consequences in critical areas like banking, healthcare, and e-commerce (Gartner, 2024). Current monitoring tools struggle with the complexity of cloud native, IoT, and edge computing environments, leaving organizations vulnerable to frequent disruptions (Brown & Green, 2025). By using an autonomous AI bot that can detect anomalies, perform automated repairs, and explain its actions, companies can lower operational costs, improve system uptime, and strengthen their defenses against cyberattacks. The explainability feature also allows IT staff to maintain oversight and confidence in automated processes, making it easier for organizations to adopt the system. Therefore, the proposed solution provides value not only through technological improvements but also by delivering real benefits in cost efficiency, reliability, and customer trust.

1.5 The conceptual, theoretical, and practical significance of the research

Conceptual Significance

This research reframes resilience in distributed systems as an active, smart, and adaptable process instead of just a backup mechanism. It introduces an autonomous AI bot that monitors, fixes, and explains its actions. This approach positions resilience as a key principle for future digital infrastructures (Kumar & Gupta, 2024; Brown & Green, 2025).

Theoretical Significance

The study presents a hybrid model that combines machine learning for spotting issues, reinforcement learning for flexible recovery, and explainable AI for clarity. This combination enhances the understanding of resilience by showing how predictive, adaptable, and clear intelligence can work together in self-healing systems (Li et al., 2025; Das & Roy, 2024; Singh & Mehta, 2024).

Practical Significance

Industries that face high costs from downtime, like cloud providers, healthcare, finance, and smart cities, can gain immediate benefits. The proposed framework not only reduces recovery time and operational costs but also builds trust in AI-driven automation by making decisions clear and reviewable (Shevchenko, 2025; Rossi & Conti, 2025; Thompson et al., 2025).

2. Literature Review

2.1 Critical Evaluation

In the last decade, there has been increasing interest in self-healing distributed systems and AI-driven resilience. Researchers have suggested using machine learning techniques to find anomalies, applying reinforcement learning for recovery decisions, and developing frameworks for cloud orchestration (Kumar & Gupta, 2024; Wang et al., 2024).

- **Anomaly Detection:** Traditional rule-based systems like Nagios and Zabbix are quick but not very flexible. Recent studies use deep learning models such as LSTMs, CNNs, and autoencoders to analyze logs and system metrics, detecting subtle anomalies more accurately. However, these models often produce a high number of false positives, making them unsuitable for real-time use (Kumar & Sharma, 2024; Li & Wang, 2025).
- **Self-Healing Strategies:** Reinforcement learning agents show potential by learning recovery policies such as container restarts and workload migration. However, most reinforcement learning systems are trained in controlled environments and find it hard to adjust to real-world distributed workloads (Li et al., 2025; Nakamura & Tanaka, 2025).
- **Explainability:** A widespread issue in the literature is the "black box problem." While AI models can identify anomalies effectively, system administrators often lack trust in automated healing due to the absence of explanations. Only a few recent studies have included Explainable AI to clarify AI-driven recovery decisions (Das & Roy, 2024; Park & Kim, 2025).

- **Commercialization:** Adoption in the industry is still limited. Most self-healing solutions remain academic prototypes due to concerns about scalability, integration with existing infrastructure, and compliance with regulations (Brown & Green, 2025; Chen & Sun, 2025).

This evaluation indicates that despite progress, current research has not fully addressed issues of trust, scalability, and practical deployment.

2.2. Research Gap

From the comparative analysis (Lit Review):

Integration Gap

Most studies focus on anomaly detection, healing, or explainability individually. Very few combine them in one framework. A complete system should include detection, decision-making, action execution, and justification to cover the entire resilience lifecycle (Kumar & Gupta, 2024; Singh & Mehta, 2024).

Trust Gap

Current self-healing systems often act like black boxes, providing little insight into their actions. Without Explainable AI (XAI), administrators may hesitate to trust automated recovery in critical situations. Including clear justifications is essential for building confidence and accountability (Das & Roy, 2024; Park & Kim, 2025).

Security Gap

Research rarely looks at adversarial machine learning attacks. Attackers can trick models into overlooking anomalies or causing false recoveries, which weakens reliability. Strong defenses are necessary to protect systems from both natural and malicious faults (Rehman & Adrian, 2025; Yoon & Lee, 2024).

Federation Gap

Most solutions assume centralized cloud systems and overlook the rise of IoT and edge environments. These situations require federated and collaborative self-healing strategies that work across distributed devices with limited resources (Rossi & Conti, 2025; Zhang et al., 2025).

Evaluation Gap

There is no standard method to test and compare self-healing systems. Many studies use custom simulations, leading to inconsistent results that are difficult to reproduce. A common testbed and evaluation framework are needed for fair comparison and industry acceptance (Shevchenko, 2025; Chen & Sun, 2025).

2.3 Comparison of Literature Review

Table 1: Lit review summary table(Key findings)

Factors	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	Propose Study
Autonomous Detection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Self-Healing Mechanism	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Adaptive Learning	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Explainable Healing Actions	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Adversarial-Aware Security	✗	✗	✓	✗	✓	✗	✗	✓	✓	✓	✓
Federated Self-Healing	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓
Human-in-the-Loop Option	✓	✗	✓	✓	✓	✓	✓	✓ Partial	✗	✓ Partial	✓
Multi-Layered Security Monitoring	✗	✗	✓	✗	✓	✓	✗	✓	✓	✓	✓
Simulation Testbed for Cyber-Attacks and Healing	✗	✓	✓	✗	✗	✓	✓	✗	✓	✗	✓
Performance Metrics and Reporting Dashboard	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 2: Lit review summary table research paper list

Group No	Articles
G1	AI-Powered Self-Healing Cloud Infrastructures: A Paradigm For Autonomous Fault Recovery Ravi Kumar Vankayalapati ¹ , Chandrashekar Pandugula ² , Venkata Krishna Azith Teja Ganti ³ , Ghatoth Mishra ⁴
G2	An Intelligent Fault Self-Healing Mechanism for Cloud AI Systems via Integration of Large Language Models and Deep Reinforcement Learning 1 st Ze Yang , 2 nd Yihong Jin, 3 rd Juntian Liu, 4th Xinhe Xu
G3	Self-Healing Networks with AI-Based Fault Prediction in IoT Ecosystems Manohar Jain
G4	AI-powered self-healing enterprise applications: A new era of autonomous systems Praveen Kumar Guguloth
G5	AI-Driven Infrastructure Automation: Leveraging AI and ML for Self-Healing and Auto-Scaling Cloud Environments Ali Asghar Mehdi Syed, Erik Anazagasty
G6	Building AI Agents for Autonomous Clouds: Challenges and Design Principles Manish Shetty ² , Yinfang Chen ³ , Gagan Somashekar ¹ , Minghua Ma ¹ , Yogesh Simmhan ⁴ , Xuchao Zhang ¹ , Jonathan Mace ⁵ , Dax Vandevoorde ⁶ , Pedro Las-Casas ¹ , Shachee Mishra Gupta ¹ , Suman Nath ⁵ , Chetan Bansal ¹ , Saravan Rajmohan ¹
G7	Artificial intelligence-enabled self-healing infrastructure systems Lauren McMillan
G8	Smart Security for Smart Clouds: Real-Time AI Defense and Self Healing Architecture Integration Asia Rehman, Grayson Adrian
G9	Self-Healing AI Systems: Autonomous Detection and Recovery from Security Breaches Holmes Walter
G10	Intelligent Cloud Resilience: Self-Healing Architectures and AI Driven Threat Automation Samiullah Khan, Nadeem Abbas

2.4 Research Objectives

2.4.1 Main Objective

The goal is to create an autonomous AI bot for distributed systems. This bot will include real time anomaly detection, self-healing capabilities, explainable decision-making, and security-aware resilience. It aims to minimize downtime, support scalability, and build trust among administrators.

2.4.2 Specific Objectives

1. We will design and implement a machine learning based anomaly detection system for distributed server environments.
2. We will integrate self-healing mechanisms using reinforcement learning, allowing the system to recover on its own from failures.
3. We will develop a module that provides clear justifications for each healing action, making these explanations understandable for humans.
4. We will include human oversight, enabling administrators to approve or override actions that carry high risk.
5. We will implement mechanisms that protect against AI-targeted attacks, ensuring resilience.

6. We will build a self-healing model that fosters collaborative resilience across distributed IoT and edge environments.
7. We will establish a simulation testbed to evaluate cyber attacks and healing in realistic situations.
8. We will create a performance dashboard that tracks detection accuracy, mean time to heal (MTTH), false positive rates, and resource usage.
9. We will assess the feasibility of commercializing the system for deployment in real world enterprise settings.

3. Research Methodology

3.1. Data Collection and Preprocessing

The success of the proposed self-healing framework relies on the quality, diversity, and representativeness of the data used to train and evaluate its components. To ensure that the anomaly detection and self-healing modules work well in real world environments, this research will use a mix of publicly available benchmark datasets and synthetic datasets created from controlled experiments in a distributed computing testbed. This combination offers both breadth and depth. Benchmark datasets capture well studied attack and anomaly patterns, while synthetic data guarantees that the system learns from faults and conditions aligned with this research's specific goals.

Benchmark Datasets

Public benchmark datasets will be used to compare with existing research. The CICDDoS2019 dataset shows traffic patterns for DoS and DDoS attacks. The UNSW-NB15 dataset includes various network intrusions such as reconnaissance, backdoors, and exploits. Open-source logs from Docker and Kubernetes clusters will capture system and application events. This will provide realistic operational data for training models to tell apart benign and malicious behaviors.

Synthetic Data Generation

To address self-healing scenarios not included in benchmark datasets, synthetic data will be created through controlled fault injection and stress tests in a Kubernetes testbed. We will simulate scenarios like CPU and memory overload, container restarts, node crashes, and network partitions. Performance logs that capture CPU, memory, I/O, and latency metrics will help models learn from real infrastructure and application failures.

Preprocessing Steps

Collected data will be cleaned, normalized, and organized for machine learning. Continuous features like CPU, memory, and I/O will be normalized. Categorical data such as error codes and container states will be encoded. Temporal patterns will be captured using sliding windows and statistical summaries. The dataset will be divided into training (70%), validation (15%), and test (15%) sets to support strong model learning and evaluation.

Importance of the Approach

Combining benchmark datasets with synthetic data and thorough preprocessing creates a realistic training set. Benchmark data offers a way to compare results, synthetic data addresses self-healing scenarios, and preprocessing ensures clean, organized inputs. This enables models to work effectively in both testing and real-world distributed systems.

3.2. Model Design and Implementation

The proposed self-healing system features a multi layer architecture that combines anomaly detection, reinforcement learning for recovery, and explainability for clear decision making. Each component is designed to operate independently while working together, forming a cohesive framework that tackles both the technical and practical issues of building resilient distributed systems.

Anomaly Detection Module

The anomaly detection module continuously checks logs, performance metrics, and network events to spot irregularities that may signal failures or attacks. Unlike traditional rule-based systems, it employs supervised and unsupervised machine learning methods. This includes LSTM networks for analyzing sequential logs, autoencoders for detecting anomalies without prior labels, and Random Forests for handling tabular data. Using ensemble learning boosts detection accuracy while keeping false positives low.

Self-Healing Module (Reinforcement Learning)

Once the system detects an anomaly, the self-healing module uses reinforcement learning (RL) to choose the best recovery action. Methods like Deep Q-Networks and Policy Gradient allow it to restart failed containers, move workloads, block harmful traffic, or allocate resources differently. The actions depend on a reward function that aims to minimize downtime, cut costs, and ensure a good user experience. Over time, the RL agent learns to autonomously select the most effective recovery strategies.

Explainability Layer

To build trust and transparency, an explainability layer offers clear insights into anomaly detections and recovery actions using explainable AI (XAI) methods such as LIME and SHAP. For instance, it shows which log features caused an anomaly alert or explains why a workload migration was needed. Summarization techniques simplify technical outputs into everyday language for administrators, who can review and approve actions before they are taken, balancing autonomy with oversight.

Integration of Components

The system combines detection, recovery, and explainability into a smooth process. The anomaly detection module activates the RL-based self-healing agent, while the explainability layer offers justifications through a dashboard. Human feedback can further refine RL policies, creating a closed-loop system that keeps improving. This integration provides an intelligent and dependable self-healing framework that tackles both technical issues and trust concerns, making it ready for real-world use.

3.3. Model Training and Evaluation

The effectiveness of the proposed self-healing system relies on both its design quality and the thoroughness of its training and evaluation. This section describes how we will train the anomaly detection models and the reinforcement learning (RL) agent. It also details the evaluation metrics and testing scenarios that will validate the system. Together, these aspects ensure the system is functional, resilient, scalable, and trustworthy in real world situations.

Training Anomaly Detection Models

We will train the anomaly detection models using historical logs and benchmark datasets such as CICDDoS2019, UNSW-NB15, and container logs. Historical logs reflect real system behaviors, while benchmark datasets offer structured attack scenarios. We will use supervised learning when labels are available and unsupervised methods like autoencoders for unlabeled data. The data will be divided into training, validation, and test sets, and we will optimize hyperparameters using grid search or Bayesian methods to improve performance and generalization.

Training the Reinforcement Learning Agent

The RL agent will learn self-healing policies by interacting with a simulated fault injection environment that reflects scenarios like server crashes, memory exhaustion, and malicious traffic. The agent will test actions such as restarting containers, migrating workloads, or blocking suspicious IP addresses. A reward function will guide learning by rewarding effective recovery and penalizing ineffective actions. We will use advanced RL algorithms like Deep Q-Networks (DQN) or Policy Gradient methods, possibly with experience replay, to ensure efficient and stable training.

Evaluation Metrics

We will evaluate the system using a mix of technical, operational, and trust metrics. We will measure anomaly detection effectiveness with accuracy, recall, and false positive rate (FPR). Mean time to heal (MTTH) will assess recovery speed. Scalability tests will evaluate performance under increasing numbers of nodes and workloads. Trust and usability will be measured through administrator surveys focusing on clarity, usefulness, and accuracy of explainability outputs.

Importance of the Evaluation Approach

This combined approach ensures that the framework is not only accurate but also reliable, scalable, and trustworthy in real-world conditions. It validates its practical use beyond just theoretical models.

3.4. Functional, Non-Functional requirements

Functional Requirements:

1. Detect anomalies in system logs and network traffic.
2. Take self-healing actions such as restarting, reallocating resources, or blocking malicious activity.
3. Provide real time monitoring through a dashboard for both web and mobile.
4. Generate explanations for healing actions that are easy for people to understand.
5. Allow administrators to override critical actions.

Non-Functional Requirements:

1. Scalability: Must support over 100 containers or nodes.
2. Reliability: At least 95% uptime in a test environment.
3. Performance: Healing latency should be under 30 seconds.
4. Security: Anomaly detection must not be bypassable through simple attacks.
5. Usability: Dashboard explanations must be rated as "clear" by at least 80% of surveyed administrators.

3.5. Specific Use of Technologies and Techniques

Functions	Technologies
DDoS Detection.	Machine Learning / Deep Learning (Scikit-learn, TensorFlow, PyTorch)
Healing Action Decision.	Reinforcement Learning (OpenAI Gym, RLlib)
Log Analysis.	Log Simulation Tools (ELK Stack / Splunk)
Self-Healing Execution.	Docker / Kubernetes
Human-readable Explanations.	LLM (ChatGPT API)
Human Admin Interface.	Flask / Node.js (Admin Approval UI)

Table 3: Specific Use of Technologies and Techniques

4. Project Timeline

Task		September				October				November				December				January				February			
		Weeks				Weeks				Weeks				Weeks				Weeks				Weeks			
		1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
1	Literature Review & Project Proposal	X	X																						
2	Proposal Presentation					X																			
3	Project design and Modeling					X	X	X	X																
4	Project development									X	X			X	X			X	X						
5	Web development													X	X	X									
6	Testing and evaluation																	X	X	X		X			
7	Project thesis																			X	X				
8	Individual report																				X				
9	For unforeseen																						X		

Table 4: Timeline

5. Experimental Plan & System Design Overview

5.1 System Design Approach

The proposed system uses an applied research approach. It focuses on developing an AI driven, autonomous self-healing bot for distributed cloud environments. The design is based on Object Oriented Analysis and Design (OOAD) principles to ensure modularity, scalability, and extensibility.

5.2 System Workflow Diagram

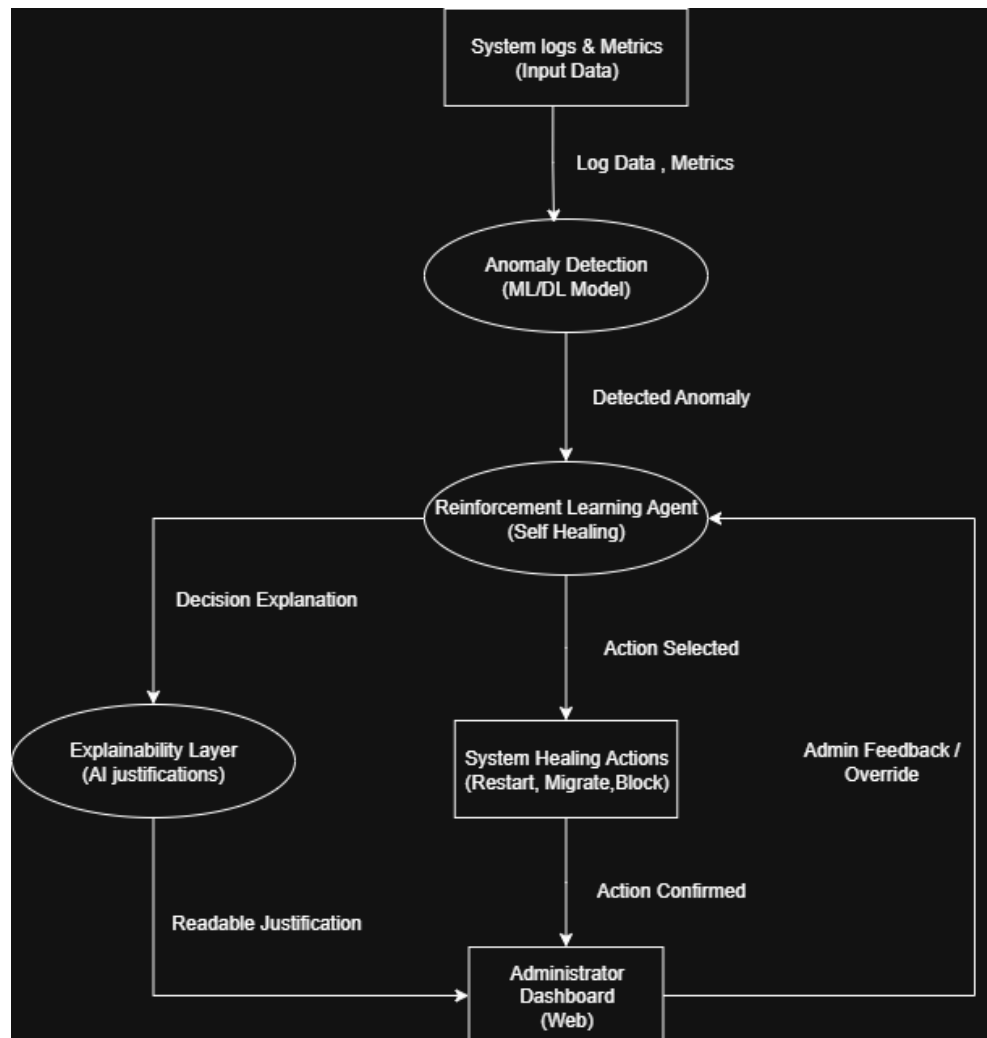


figure 1: System Workflow Diagram

5.3 System Architecture Diagram

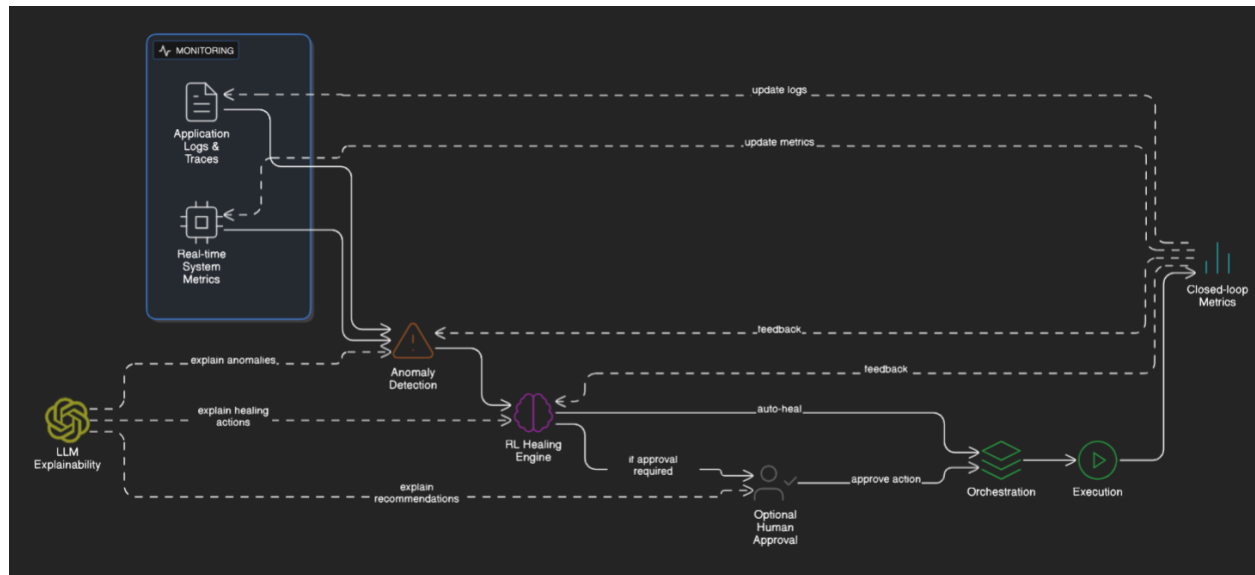


figure 2: proposed Architecture of autonomous AI bot for self-healing in distributed systems

5.4 Materials Required

Datasets

High-quality datasets are essential for training the self-healing system. The CICDDoS2019 dataset provides labeled traffic data for DoS and DDoS attacks. This enables the system to distinguish between normal and malicious traffic. The UNSW-NB15 dataset includes over two million records covering various types of network intrusions. This ensures the anomaly detection module can handle diverse attack scenarios. Additionally, synthetic datasets will be created using Kubernetes fault injection and resource stress tests. These will simulate realistic failures like node crashes, container restarts, and resource exhaustion, which are not always captured in public datasets.

Logs and Performance Metrics

System and application logs from Docker and Kubernetes will be collected to monitor runtime events, errors, and system states. Alongside logs, performance metrics such as CPU usage, memory consumption, I/O throughput, and latency will be tracked in real-time. These metrics help the system detect both static anomalies from datasets and dynamic irregularities in live operations. This ensures robust detection and recovery.

Survey Instruments

To evaluate trust and usability, structured survey instruments will be used. These surveys will include Likert-scale questions to measure administrator perceptions of the system's transparency, clarity of AI-generated explanations, and willingness to trust automated recovery decisions. This human-centered evaluation complements technical testing and shows real-world acceptability.

Hardware Requirements

The development environment will include a workstation with Intel i7 or AMD Ryzen processors, 16 to 32 GB RAM, and 1 TB storage for coding, debugging, and initial model testing. A testbed cluster of 3 to 5 physical or virtual servers will host Docker and Kubernetes environments for fault injection and anomaly recovery experiments. One of these servers will include a GPU such as the NVIDIA RTX 3060 or higher to accelerate deep learning model training, especially for LSTM networks and autoencoders.

Networking Equipment

High-speed LAN or WAN will connect the testbed servers to simulate distributed environments realistically. This setup allows testing of network-related anomalies, including changes in bandwidth, latency, packet loss, and congestion. It provides a realistic scenario for evaluating system performance, resilience, and scalability.

6. Conclusion

This research proposal outlines the design of an autonomous AI bot that combines anomaly detection, reinforcement learning-based self-healing, and explainable AI to improve resilience in distributed systems. By addressing important research gaps like integration, trust, security, federation, and evaluation, the study offers a unified framework that moves beyond traditional monitoring methods.

The proposed system aims to reduce downtime, shorten recovery times, and build administrator trust through clear decision-making. While challenges remain, such as handling false positives and simplifying technical explanations, the framework offers a strong foundation for both academic research and practical use.

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8.Appendices

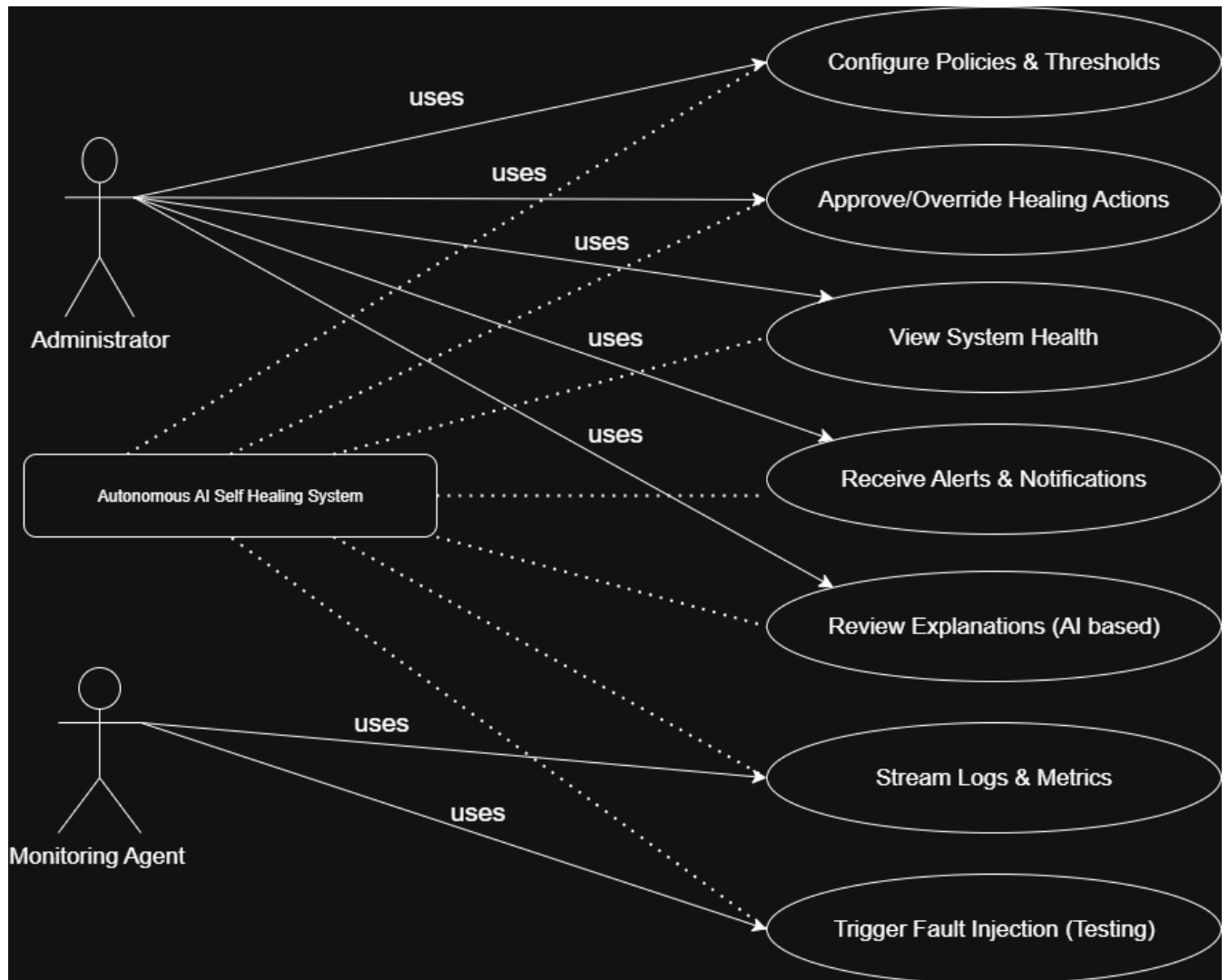


figure 3: UML Use Case Diagram